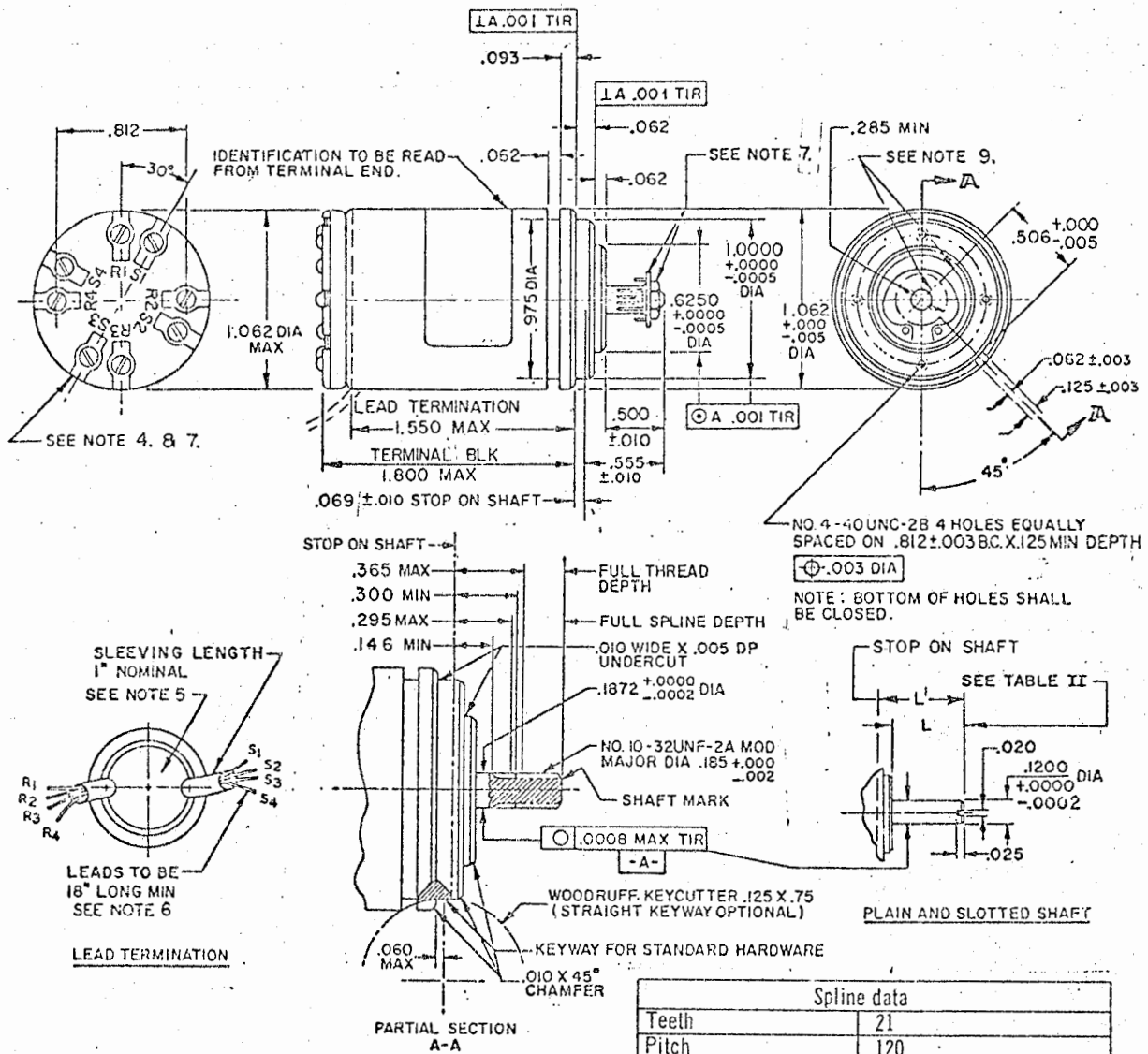
MILITARY SPECIFICATION SHEET

RESOLVER, ELECTRICAL

TYPE 11R1N4a

The complete requirements for procuring the resolver described herein shall consist of this document and the issue in effect of MIL-R-23417.



Spline data	
Teeth	21
Pitch	120
Pitch diameter	.175 +.000, -.002
Outside diameter	.1872 +.0000, -.0005
Pressure angle	20°
Root diameter maximum	.155
Tooth form	Full depth involute

Note: The end function of this spline is for positive rotational fastenings. It is not to be used as a gear. This spline shall be inspected by (spline) go composite ring gage MS 90406.

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 FSC 5990

NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerances on decimals are .XXX ±.005 and angles ±2°.
3. Perpendicularity and concentricity shall be measured with the unit in a vertical position supported by the shaft, then the housing rotated.
4. Terminal sequency shall be shown and identified by molded lettering on the end cap or permenently affixed thereto. The end cap and terminal block shall be secured by screws other than the terminal screws. Mounting screw positions optional. Terminal block designed for MS 17182 terminal lug (an approved 12-terminal block may be used provided all unused terminal positions are closed).
5. Minor variations of unit configuration are permitted for undimensioned design detail.
6. Leads shall conform to MIL-W-16878/4, type E-28.
7. The following list of hardware shall be packaged in a ventilated envelope made of material. having a PH value of 6-8 and compatable with the weight of the contents. The packaged hardware shall be placed in the unit resolver as follows:

Machine screw - - - - - - -	(MS 35276-213)	8 Required	(*) (**)
Lockwasher - - - - - - - -	(MS 35338-134)	8 Required	(*) (**)
Drivewasher - - - - - - - -	(MS 17186-6)	1 Required	(*)
Nut - - - - - - - - - - - - -	(MS 17187-2)	1 Required	(*)

(*) For units with spline shaft and terminal board termination.
(**) For units with plain smooth shaft and terminal board termination.

8. For information only: Use mounting clamp MS 17183 or equal.
9. Mark on housing is to match shaft mark within 10° when rotor is set at approximately resolver zero, and will bear no relation to the center lines shown on the drawing.
10. A voltage value of .5 to 26 volts shall be entered in the 'volts' designation space on the identification plate.
11. The weight shall be attached to the end of each wire, pulling straight away from the resolver. The lead shall then be bent 90° at the point of exit from the housing and while in this attitude the resolver rotated 360° counterclockwise. Any indication of the lead pulling loose, wire strands breaking, or permanent deformation of the insulation shall be cause for rejection under qualification tests.
12. Pending issuance of a qualified products list (QPL) a sample of the resolvers to be furnished under this specification shall be subjected to the pre-production tests as specified in MIL-R-23417.

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TABLE I

RESOLVER, ELECTRICAL: TYPE 11R1N4a

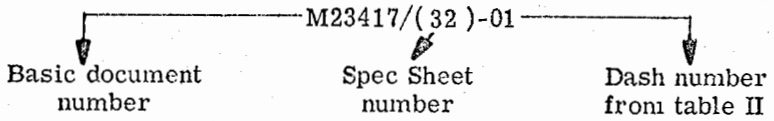
PERFORMANCE REQUIREMENTS AND CHARACTERISTICS

Requirements	Value	Unit	Tolerance
1. Excitation winding	Stator	---	---
2. Rated frequency	400	Hz	±1%
3. Input voltage:	11.8	Volts rms	±1%
Test	11.8	Volts rms	±1%
4. Primary current	143	Milliamps	Nominal
5. Primary power	0.25	Watts	Nominal
6. DC resistance:			
Rotor	19	Ohms	±15%
Stator	7.8	Ohms	±15%
7. Impedance:			
Zro	28.5 +j106	Ohms	R±15%, +jx±10%
Zso	16.2 +j93.6	Ohms	R±15%, +jx±10%
Zrs	28 +j11 (Reference)	Ohms	---
8. Transformation ratio			
Output/input	1.000	---	±2%
9. Equality of transformation ratio			
Output/input	±0.1	Percent	---
10. Phase shift			
Output/input	5 (Lead)	Degrees	±1.0
11. Accuracy			
Electrical error	±3.0	Minutes	Maximum
12. Null voltage:			
Fundamental	15	Mv rms	Maximum
Total	20	Mv rms	Maximum
13. Shift of resolver zero			
With frequency	1	Minutes	Maximum
14. Temperature rise	10	°C	Maximum
15. Friction torque:	0.07	Oz-In	Maximum
@ -55°C	0.15	Oz-In	Maximum
16. Radial play	0.0006 (For reversal of 1/2 lb load)	Inches	Maximum
17. End play	0.0005 - 0.0015 (For reversal of 1 lb load)	Inches	Min-Max
18. Wire lead stress (see note 11)	2	Pound	Minimum

Intended use: This resolver is intended primarily for use as a resolver differential (RD) in a data transmission system; e.g., a RX-RD-RC chain.

Part number: The part number consists of the basic number of this specification sheet and a dash number taken from table II.

Example:



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TABLE II

Dash no.	Shaft diameter	Shaft style	Termination	Shaft length	
				L (±.010)	L' - Dim to stop on shaft (±.010)
-01	.187	Spline	Terminal board	.500 (1/2)	.555
-02	.120	Plain	Wire leads	.312 (5/16)	.367
-03	.120	Plain	Wire leads	.375 (3/8)	.430
-04	.120	Plain	Wire leads	.437 (7/16)	.492
-05	.120	Plain	Wire leads	.500 (1/2)	.555
-06	.120	Plain	Wire leads	.562 (9/16)	.617

Custodians:
Army - MU
Navy - AS
Air Force - 11

Review activities:
Army - MU, EL
Navy - AS, SH
Air Force - 11, 17, 85

User activities:
Army - MI, ME, AV
Navy - CG, MC
Air Force - 19

Preparing activity:
Air Force - 11

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